ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY
A.S Nagar, Elupatti– Thanjavur.
(Approved by AICTE-New Delhi, Affiliated to Anna University, Chennai.)

Department of Mechanical Engineering

CONTINUOUS INTERNAL ASSESSMENT TEST-II, OCT - 2026

Reg.No	8	2	1	9								
Regulation	**R2025**											
Course Code-Name	**ME25C03 & INTRODUCTION TO MECHANICAL ENGINEERING**								**SET- B**			
Year / Semester	**I & I**								**Total Marks:100**			
Date - Time	**23. 10. 2026 / 9.30 to 12.30**											
Course Instructor	**Mr. B.Nagendran AP/MECH**											

C102.1 – Explain core mechanical engineering concepts.

C102.2 – Apply basic engineering calculations in mechanical engineering.

L1-Remember , L2-Understand , L3-Apply , L4-Analyze , L5 -Evaluate , L6-Create

Answer All Questions
PART – A (10 × 1 = 10 Marks & 10 × 3 = 30 Marks)

Q.NO	QUESTIONS	Marks	CO	LEVEL
1.	Work is mathematically expressed as the product of: a) Mass and acceleration b) Force and displacement c) Power and velocity d) Energy and mass	**1**	C102.1	**L2**
2.	Power is defined as: a) The ability to do work b) The change in position c) The rate of doing work d) The product of energy and mass	**1**	C102.1	**L2**
3.	The energy associated with the motion of an object is called: a) Potential energy b) Thermal energy c) Kinetic energy d) Chemical energy	**1**	C102.1	**L2**
4.	Which of the following is not an SI unit? a) Kilogram b) Metre c) Litre d) Kelvin	**1**	C102.1	**L2**
5.	In the formula for weight (W = mg), what value is typically used for g? a) 10 m/s b) 9.81 m/s² c) 1.6 m/s² d) 3.14 m/s²	**1**	C102.1	**L1**
6.	Which branch of science describes the state of rest or motion of bodies under external loads? a) Thermodynamics b) Fluid mechanics c) Mechanics of materials d) Kinematics	**1**	C102.2	**L1**
7.	Mechanics of rigid bodies is further divided into: a) Statics and dynamics b) Elasticity and plasticity c) Stress and strain d) Tension and compression	**1**	C102.2	**L2**
8.	Which of the following deals specifically with bodies at rest? a) Dynamics b) Statics c) Kinetics d) Thermodynamics	**1**	C102.2	**L1**

9.	The internal resistance offered by a material against deformation is known as: a) Load b) Stress c) Strain d) Elasticity	**1**	C102.2	**L1**
10.	Strain is defined as the ratio of: a) Load to area b) Original dimension to change in dimension c) Change in dimension to original dimension d) Stress to Young's modulus	**1**	C102.2	**L1**
11.	Discuss the types of motion.	**3**	C102.1	**L1**
12.	Define pressure and torque.	**3**	C102.1	**L1**
13.	What are the fundamental (base) units in the SI system?	**3**	C102.1	**L1**
14.	Define engineering ethics. What is its role in engineering?	**3**	C102.1	**L1**
15.	Describe an engineer's responsibility regarding whistleblowing.	**3**	C102.1	**L1**
16.	Explain the elastic limit and the plastic region.	**3**	C102.2	**L1**
17.	Distinguish between ductility and brittleness.	**3**	C102.2	**L2**
18.	Explain tensile and compressive stresses.	**3**	C102.2	**L1**
19.	Define a beam and state its function.	**3**	C102.2	**L1**
20.	Compare a cantilever beam and a simply supported beam.	**3**	C102.2	**L1**
PART – B (5 × 12 = 60 Marks)				
21.a	Explain in detail the ethics and professionalism in engineering.	**12**	C102.1	**L3**
(OR)				
21.b	Explain the types of motion with examples.	**12**	C102.1	**L3**
22.a	Discuss in detail the derived units in the SI system.	**12**	C102.1	**L3**
(OR)				
22.b	Discuss the historical transition from the steam engine to the internal combustion engine and the engineering challenges involved.	**12**	C102.1	**L3**
23.a	Explain the following: (i) Hooke's law (ii) Young's modulus (iii) Modulus of rigidity (iv) Bulk modulus.	**12**	C102.1	**L3**
(OR)				
23.b	Explain tensile, compressive and shear stresses with suitable diagrams and units.	**12**	C102.1	**L3**
24.a	Define and explain the mechanical properties of materials.	**12**	C102.2	**L3**
(OR)				
24.b	Explain the different types of trusses.	**12**	C102.2	**L3**
25.a	Explain the types of stresses.	**12**	C102.2	**L3**
(OR)				
25.b	A cantilever beam of length 3 m carries a uniformly distributed load of 1000 N/m over the whole span. Find the reaction and the fixing moment at the fixed support.	**12**	C102.2	**L4**

Course Instructor **HOD** **IQAC Coordinator**